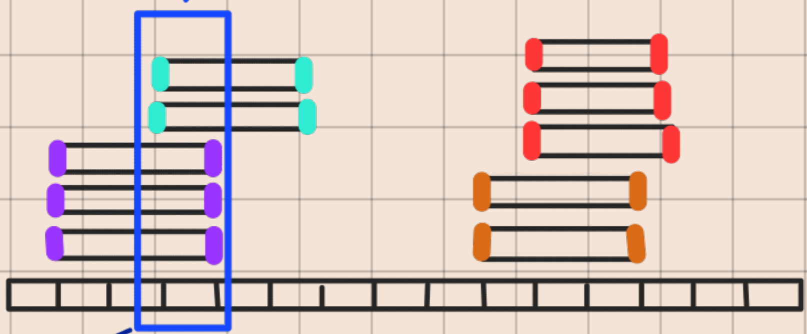
First, iterate over the loci

Next, iterate over the reads at that locus.



pileup at the current locus:

UMI 1 $\left\{ \begin{array}{c} A \\ A \\ \vdots \\ A \end{array} \right.$

UMI 2 $\left\{ \begin{array}{c} A \\ T \\ \vdots \\ A \end{array} \right.$

$$\varepsilon_{UMI\,1} = \frac{\#\text{ non major alleles}^{*}}{\#\text{ reads with UMI 1}}$$

$$w_{UMI\,1} = \#\text{ reads with UMI 1}$$

$$\varepsilon_{UMI\,2} = \frac{\#\text{ non-major alleles}^{*}}{\#\text{ reads with UMI 2}}$$

$$w_{UMI\,2} = \#\text{ reads with UMI 2}$$

Finally, calculate the weighted average error rate per locus.

$$\varepsilon_{LOCUS\,x} = \text{wmean}\left(\varepsilon_{UMI\,1} * w_{UMI\,1},\ \varepsilon_{UMI\,2} * w_{UMI\,2}\right)$$

* The major allele is just the most common allele.
The non-major alleles are the rest.

$$\left.\begin{matrix}A\\A\\A\\T\\A\\G\\A\end{matrix}\right] \begin{matrix}\rightarrow \text{major} = A\\ \rightarrow \text{n. non-major} = 2\\ \rightarrow \varepsilon = 2/7\end{matrix}$$